

FEASIBILITY ASSESSMENT & GO/NO-GO EVALUATION

EEG-Based Clinical Decision Support Tool

Product Feasibility & Business Analysis Study

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1. Purpose of Assessment

This document evaluates the feasibility of an AI-assisted clinical decision support concept for schizophrenia risk classification using EEG-derived neurophysiological data and machine learning techniques.

The assessment analyzes the project across five dimensions — technical, clinical, operational, regulatory, and ethical — and concludes with a formal go/no-go recommendation based on evidence gathered during the proof-of-concept study. The objective is to determine whether continued exploration and future investment in this concept are justified.

2. Project Summary

The feasibility study evaluated machine learning classification capability using EEG-derived NREM sleep biomarkers from 130 subjects and approximately 2,894 healthcare features.

Analytical Workflow

- Leakage-safe preprocessing pipeline (median imputation, StandardScaler, OneHotEncoding)
- Feature selection using ANOVA SelectKBest (K=100)
- Stratified 80/20 train-test split (104 train / 26 test)
- 5-fold stratified cross-validation
- Comparative ML model evaluation across four algorithms

Models Evaluated & Results

Model	Test Accuracy	Test AUC	CV AUC (Mean ± Std)
SVM (Best Model)	84.6%	0.896	0.835 ± 0.105
Random Forest	80.8%	0.884	0.845 ± 0.091
Logistic Regression	80.8%	0.875	—
Decision Tree	65.4%	0.649	—

3. Feasibility Assessment

3.1 Technical Feasibility

Assessment: PROMISING

The study demonstrated meaningful classification capability using EEG-derived neurophysiological features. The SVM model exceeded the predefined AUC threshold of 0.80, and the leakage-safe preprocessing pipeline maintained methodological rigor throughout.

Key Findings

- SVM achieved ROC-AUC 0.896 — above the predefined success threshold of 0.80
- Fast spindle density features (DENS_15) across frontal and central channels showed the strongest predictive contribution, consistent with published neuroscience literature
- Leakage-safe preprocessing and stratified cross-validation maintained methodological integrity
- Random Forest CV AUC of 0.845 ± 0.091 confirms multi-model consistency

Technical Limitations

- Small test set of 26 samples — a single misclassification shifts accuracy by approximately 4%
- CV standard deviation of ± 0.105 (SVM) indicates fold-level instability
- No hyperparameter tuning performed — default model parameters used throughout
- Pre-extracted EEG features only — raw EEG signal processing pipeline not implemented
- No external validation dataset available

Conclusion: Technical feasibility is considered promising for continued exploratory research. Additional validation with larger datasets is mandatory before any deployment consideration.

3.2 Clinical Feasibility**Assessment: CONDITIONAL**

The workflow concept aligns with existing psychiatric evaluation processes. NREM sleep EEG biomarkers are established in peer-reviewed literature as neurophysiological markers of schizophrenia, providing a validated scientific foundation for the proposed decision-support layer.

Clinical Strengths

- Biomarker foundation validated by Kozhemiako et al. (2022, eLife) - replicable across Wave 1 and Wave 2 independent cohorts
- As-Is to To-Be workflow transition designed to supplement — not replace — clinician judgment
- Potential for earlier risk stratification before full symptom onset
- Structured decision-support output reduces diagnostic variability

Clinical Constraints

- No prospective clinical validation study conducted
- Model trained on research-grade EEG data — clinical acquisition environments introduce additional noise
- Diagnostic accuracy required for clinical use significantly exceeds the current feasibility study standard
- Clinician adoption, training requirements, and change management have not yet been assessed
- AI outputs are not sufficient for standalone diagnosis — psychiatric evaluation remains clinician-led

Conclusion: Clinical feasibility is conditional and dependent upon prospective validation within real-world psychiatric environments before any integration consideration.

3.3 Operational Feasibility

Assessment: **CONDITIONAL**

The proposed workflow could be integrated into existing healthcare operations through an EHR-compatible decision-support architecture. Established standards such as HL7 FHIR provide a viable integration pathway for major EHR platforms.

Operational Strengths

- EHR integration feasible via HL7 FHIR standard — supported by Epic, Cerner, and equivalent platforms
- Standardized ML pipeline and reproducible preprocessing support operational consistency
- Role-based access control and audit trail requirements identified
- Cloud and on-premises deployment architectures are both viable

Operational Constraints

- EEG acquisition infrastructure is not universally available across clinical settings
- Standardized NREM sleep EEG capture protocols would need to be established at deployment sites
- Staff training requirements for EEG acquisition and CDSS output interpretation were not assessed
- Integration costs, implementation timeline, and resource requirements have not been estimated

Conclusion: Operational feasibility is conditional and would require institution-specific infrastructure planning and workflow integration analysis before deployment.

3.4 Regulatory Feasibility

Assessment: **NOT READY**

Healthcare AI systems intended for clinical decision support fall under significant regulatory oversight. The proposed system would be classified as Software as a Medical Device (SaMD) under FDA guidance, requiring pre-market submission before any clinical deployment.

Regulatory Pathway

- Likely Class II SaMD classification under FDA guidance
- Class II SaMD requires 510(k) pre-market notification demonstrating substantial equivalence to a predicate device
- No established predicate device for EEG-based schizophrenia risk classification currently exists — De Novo pathway may be applicable
- HIPAA compliance required for all patient data handling, storage, and transmission
- IRB approval required for any clinical data collection or human subjects' research

Current Regulatory Status

- Project remains exploratory, analytical, and non-deployable in its current form
- No regulatory submission or clinical certification process initiated
- No clinical validation data package available for submission
- Post-market surveillance plan not yet defined

Conclusion: Regulatory feasibility is currently not sufficient for deployment consideration. FDA SaMD pathway identified; significant pre-submission work required, including clinical validation and regulatory consultation.

3.5 Ethical & Responsible AI Feasibility

Assessment: **CONDITIONAL**

The project acknowledges the importance of responsible AI governance in psychiatric healthcare applications. The system is explicitly designed as decision-support only, with clinician authority maintained throughout the diagnostic process.

Ethical Strengths

- AI positioned strictly as decision-support — final diagnosis remains with treating psychiatrist
- Anonymized research dataset used — no real patient data involved in feasibility study
- Feature importance scores (ANOVA-based) provide basic analytical transparency
- Transparent limitation documentation included across all project artifacts

Ethical Constraints

- Dataset skewed toward the 25–45 age range and 64% male — limited demographic representation
- Model performance across diverse demographics (age, sex, ethnicity) has not been validated
- No formal bias audit conducted
- A full explainability framework beyond ANOVA feature importance is not implemented
- Patient consent framework for AI-assisted psychiatric screening is not defined

Conclusion: Ethical feasibility is conditional upon implementation of a bias audit, diverse validation dataset, and responsible AI governance framework before any clinical deployment.

4. Go / No-Go Evaluation Framework

Criteria	Weight	Status	Decision
ML model achieves AUC ≥ 0.80	High	☑ Met — SVM AUC 0.896	☑ Go
Leakage-safe, reproducible pipeline	High	☑ Met — stratified CV, no leakage	☑ Go
Biomarkers validated in the literature	High	☑ Met — Kozhemiako et al. 2022	☑ Go
Sample size adequate for clinical validation	High	✗ Not Met — 130 subjects insufficient	✗ No-Go
External validation dataset available	High	✗ Not Met — single cohort only	✗ No-Go
Regulatory pathway clear and ready	High	⚠ Partial — pathway identified, not ready	⚠ Conditional

Criteria	Weight	Status	Decision
EHR integration infrastructure confirmed	Medium	⚠️ Partial — conceptually viable, not confirmed	⚠️ Conditional
Bias and fairness audit completed	High	❌ Not Met — not conducted	❌ No-Go
Explainability framework implemented	Medium	⚠️ Partial — ANOVA importance only	⚠️ Conditional
Clinical validation study completed	High	❌ Not Met — not conducted	❌ No-Go

5. Overall Go / No-Go Recommendation

Decision: CONDITIONAL GO
Go on with continued research and development.
No-Go on clinical deployment in current form.

The feasibility study produced technically valid and scientifically grounded results. The SVM model achieved AUC 0.896 with a leakage-safe pipeline, and the biomarker foundation is supported by peer-reviewed literature. These results justify continued investment and development.

However, four high-weight criteria remain unmet — sample size, external validation, bias audit, and clinical validation study. These are not minor gaps; they are fundamental requirements for any healthcare AI system progressing toward clinical use.

6. Conditions for Full Go

Before a full Go decision on clinical deployment can be made, the following must be completed:

1. Expand the dataset to a substantially larger dataset across diverse demographics (age, sex, ethnicity)
2. Conduct external validation on at least one independent clinical cohort
3. Complete a formal bias and fairness audit across all demographic subgroups
4. Conduct a prospective clinical validation study in a real psychiatric clinical environment
5. Identify regulatory pathway (510(k) or De Novo) and initiate FDA consultation
6. Implement a full explainability framework for clinical output transparency
7. Confirm EHR integration feasibility at target deployment sites
8. Develop HIPAA-compliant data infrastructure and IRB approval framework

7. Conclusion

This project successfully demonstrated the value of combining machine learning validation, healthcare analytics, and business analysis to evaluate an AI-assisted clinical decision support concept. The structured feasibility framework applied here provides a replicable methodology for evaluating future healthcare AI initiatives.

While the current implementation is not suitable for real-world clinical deployment, the study establishes a credible proof-of-concept foundation. The technical results are promising, biomarker science is grounded in published research, and the business analysis artifacts define a clear pathway from concept to deployment readiness.

The recommendation is to continue development under the conditions outlined in Section 6, to progress from feasibility demonstration to a clinically validated, regulatory-aligned decision support system.

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